

MOHAMED ABO EL-SEOUD

Backend Engineer Cairo, Egypt

+201007045993

linkedin.com/in/mohamed-aboelseoud

MAES-Pyramids

mohamedaboelseoudatia@gmail.com

SUMMARY

Backend Engineer with 2+ years of experience building scalable systems across **proptech**, **social media**, and **e-commerce** domains. Specialized in designing microservices, optimizing performance through caching and asynchronous pipelines, and managing cloud infrastructure on a daily basis. Focused on system reliability, clean architecture, and shipping production-grade solutions.

SKILLS

Tech Stack: TypeScript, Go, SQL, GraphQL, NestJS, Express, React, Next.js, PostgreSQL, MongoDB, Redis Search, BullMQ.

Tools & Systems: Docker, AWS, Terraform, n8n, Datadog, NewRelic, OpenSearch, Sentry, Mixpanel, MoEngage, LangChain.

EXPERIENCE

Apr 2024 – Present

Full-Stack Engineer at Nawy

Cairo

- Contributed to the development of a **full-lifecycle CRM platform** supporting complex **multi-organization**, **multi-funnel** lead management with configurable **auto-assignment engines**, **lead delegation hierarchies**, and **agent performance rating**, enabling structured and traceable sales operations.
- Reduced **database CPU utilization by ~32%** (from ~22% to ~15%) through two complementary strategies: introduced **Redis Search** for hierarchy-intensive queries — backed by **BullMQ** for async synchronization and **AWS Lambda** for continuous accuracy monitoring — and applied repository-wide **Redis caching** with dynamic cache invalidation across lookup paths.
- Built an **event aggregation and enrichment pipeline** that collects and unifies user activity from multiple providers using cohort syncing, leveraging **LLMs** to infer **user intent** and trigger **agent-driven actions**. Increased utilization of **previously inactive and non-target customer segments**, while managing **rate limits**, **LLM costs**.
- Contributed to AI-driven features across the system, including automated **task quality evaluation** via **LLMs** and **LangChain**, **lookup localization** through structured prompt templates, and **multilingual audio transcription** powered by **OpenAI Whisper**.
- Built **n8n workflows** for the auto-assignment service revamp, replacing hardcoded logic with **visual rule configuration** and conditional branching accessible to non-technical stakeholders.
- Contributed to AWS infrastructure setup using **Terraform** (ECS, API Gateway, Cognito, RDS), including **pre-token generation Lambdas**, **custom authorization flows**, and **authorization and proxy Lambdas** for secure third-party integrations, aligned with AWS security best practices.

NestJS / PostgreSQL / Redis / AWS / Terraform / n8n / NewRelic / Datadog

Aug 2023 – Apr 2024

Backend Engineer at Baianat

New Damietta

- Delivered backend systems across **e-commerce**, **catering**, **e-learning**, and **social media** domains using **NestJS**, **GraphQL**, **PostgreSQL**, **MongoDB**, and **AWS**, with consistent patterns for authentication, rate limiting, and error handling.
- Built real-time features including **live chat** with **Firebase push notifications** and **GraphQL Subscriptions**, and shipping workflows with geographic zone support.
- Implemented secure content delivery via **S3 presigned URLs** and **Bunny CDN**, optimized full-text search, and async job processing using **BullMQ/Redis** queues.

NestJS / GraphQL / PostgreSQL / MongoDB / Redis / AWS

Mar 2023 – Jul 2023

Backend Intern at Tuwaiq

El Sheikh Zayed

Built and maintained **RESTful APIs** with comprehensive documentation, applying consistent patterns for **authentication**, **validation**, and **error handling**.

PROJECTS

Sakan

Freelance – Sep 2025 – Feb 2026

A marriage matching platform with full-stack ownership across frontend, backend, and deployment.

- Built **Micro-Frontend Architecture** using **Next.js** and **Chakra UI**.
- Engineered **Automated Matching Pipelines** with **AI-Powered Chat Monitoring**.
- Incorporated **Moyasar** for subscription-based payments and **Twilio** for messaging.
- Deployed on **Alibaba Cloud** with dedicated **Worker Services** for matching, mail, and **SSE**-based real-time updates.

Next.js / NestJS / OpenAI API / Alibaba Cloud / Twilio

FRA Sandbox

Baianat – Nov 2024 – Feb 2025

A fintech sandbox for **Egypt's Financial Supervisory Authority** to drive innovation.

- Built **Multi-Role Authentication** supporting innovators and enablers (accelerators, NBFIs, VCs).
- Developed **Hackathon Management** with team discovery, role-based members, and invitation flows.
- Established **Real-Time Discussion Boards** with comments, replies, and notifications.

- Created **Live Ticketing System** for real-time support and issue tracking.

NestJS / PostgreSQL / TypeORM / GraphQL Subscriptions

RIGOW

Baianat – Sep 2024 – Nov 2024

A social media platform for athletes and sports enthusiasts.

- Built **Real-Time Chat** with message status tracking (sent, delivered, edited).
- Crafted **AI-Driven Timeline** using **TensorFlow** for content recommendations.
- Established **Ranking System** prioritizing content based on user engagement.
- Applied typo-tolerant **Full-Text Search** for exploring topics and profiles.
- Created **Notification System** with localized alerts and optimized batch delivery.
- Enabled **Social Auth** via OAuth2 OpenID for Facebook, Google, and Apple.
- Used **Redis Pub/Sub** for real-time **GraphQL Subscriptions**.

NestJS / Redis / GraphQL / TensorFlow / PostgreSQL

FASILA

Freelance – Jan 2024 – Mar 2024

An e-learning platform providing university students with affordable access to lectures and resources.

- Utilized **Worker Threads** to optimize CPU-intensive tasks.
- Ensured **Data Integrity** using **MongoDB Transactions** for purchases.
- Connected **Paymob** for seamless payments (card & wallet).
- Enforced **Single-Session Restrictions** to prevent account sharing.
- Secured content using **Presigned URLs**, CORS, and encryption.
- Configured **WhatsApp API** for cost-effective OTP delivery.
- Deployed **Dockerized services** on **DigitalOcean** with **Nginx Load Balancing** and **PM2**.

Express / MongoDB / Redis / Docker / DigitalOcean

WORKFLOW

- **Performance Analysis:** Used **Datadog** and **NewRelic** to identify intensive endpoints, analyze slow database queries, detect lock contention, and drive targeted optimizations.
- **AI-Assisted Workflow:** Leveraged **Claude** and **Cursor** for accelerated engineering, including generating project rules, custom skills, and **MCP** tool integrations such as **Figma** and **Playwright** to streamline code generation, review cycles, and end-to-end testing.
- **Cloud & Infrastructure:** Worked daily with **AWS** services including **Cognito**, **SNS**, **SQS**, **ECS**, **API Gateway**, **Lambda**, **Secrets Manager**, and **RDS**. Used **Terraform** to provision Lambda functions and API Gateway configurations. Managed containerized deployments with **Docker**.

EDUCATION

B.Sc in Computer Science

Mansoura University

Offered a **Teaching Assistant** position — declined to pursue industry.

GPA: **91.1%**

Sep 2020 – Jul 2024

LANGUAGES

English - Professional, Arabic - Native

CERTIFICATIONS

- **NTI:** Web Development (Node & React)
- **ITI:** Backend Development (Laravel & PHP)
- **CNBC:** 90% Most Common AWS Services
- **Udemy:** Troubleshooting Backend Performance